

Day 3 – Raw OpenRocket Simulation Data

Baseline Simulations

Trial	Motor	Vel off Rod (m/s)	Apogee (m)	Opt Delay (s)	Max Vel (m/s)	Max Accel (m/s²)	Time to Apogee (s)	Flight Time (s)
Trial 1	C6-5	21.0	54.0	1.29	85.6	304	3.15	7.26
Trial 2	C6-5	20.6	193.0	3.24	83.2	300	5.10	14.90
Trial 3	C6-5	20.9	51.6	1.23	84.7	304	3.09	7.09
Trial 4	C6-5	20.9	50.2	1.23	84.3	303	3.04	7.00
Trial 5	C6-5	20.9	49.1	1.21	83.7	303	3.02	6.94

Modified Simulations

Trial	Motor	Vel off Rod (m/s)	Apogee (m)	Opt Delay (s)	Max Vel (m/s)	Max Accel (m/s²)	Time to Apogee (s)	Flight Time (s)
Trial 1	C6-5	21.0	53.7	1.31	85.5	304	3.12	7.21
Trial 2	C6-5	21.0	52.6	1.28	84.9	304	3.09	7.15
Trial 3a	C6-5	20.9	51.6	1.27	84.6	304	3.08	7.11
Trial 3b	C6-5	20.9	51.4	1.23	84.7	304	3.09	7.06
Trial 4	C6-5	20.9	49.8	1.20	83.9	303	3.06	6.96
Trial 5	C6-5	20.9	51.9	1.25	84.3	304	3.11	7.13